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Refrigeration and Air-Conditioning

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Syllabus

Module I

Principles of refrigeration-unit of refrigeration - capacity - Coefficient of Performance - refrigeration systems: Carnot refrigeration cycle - Steam jet refrigeration - Thermoelectric refrigeration - vortex tube - pulse tube - air refrigeration cycle boot strap & boot strap evaporating cooling - thermodynamic analysis of Bell- Coleman cycle.

Module II

Vapour compression system - theoretical and practical cycles - simple and multi pressure systems - thermodynamic analysis - vapour absorption system - principle of operation of aqua - ammonia and lithium bromide - water systems - Electrolux system - comparison between vapour compression and absorption systems - refrigerants - thermodynamic, physical and chemical properties of refrigerants, environment friendly refrigerants and its properties, selection criteria of refrigerants.

Refrigerants

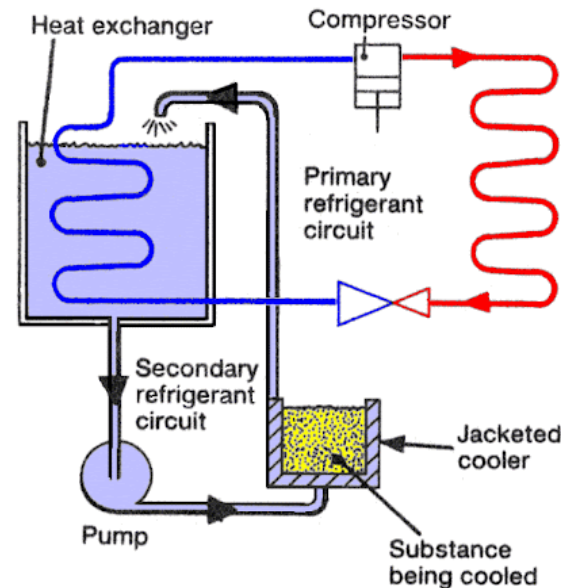
Refrigerants



- Refrigerants:- Heat carrying medium in refrigeration cycle
- Absorb heat from low temperature medium
- Reject heat to higher temperature medium
- First refrigerant:- Natural Ice, mixture of ice and salt
- Suitability of a refrigerant for a certain application is determined by its physical, thermodynamic, chemical properties and by various practical factors
- There is no one refrigerant which can be used for all types of applications i.e., there is no ideal refrigerant.

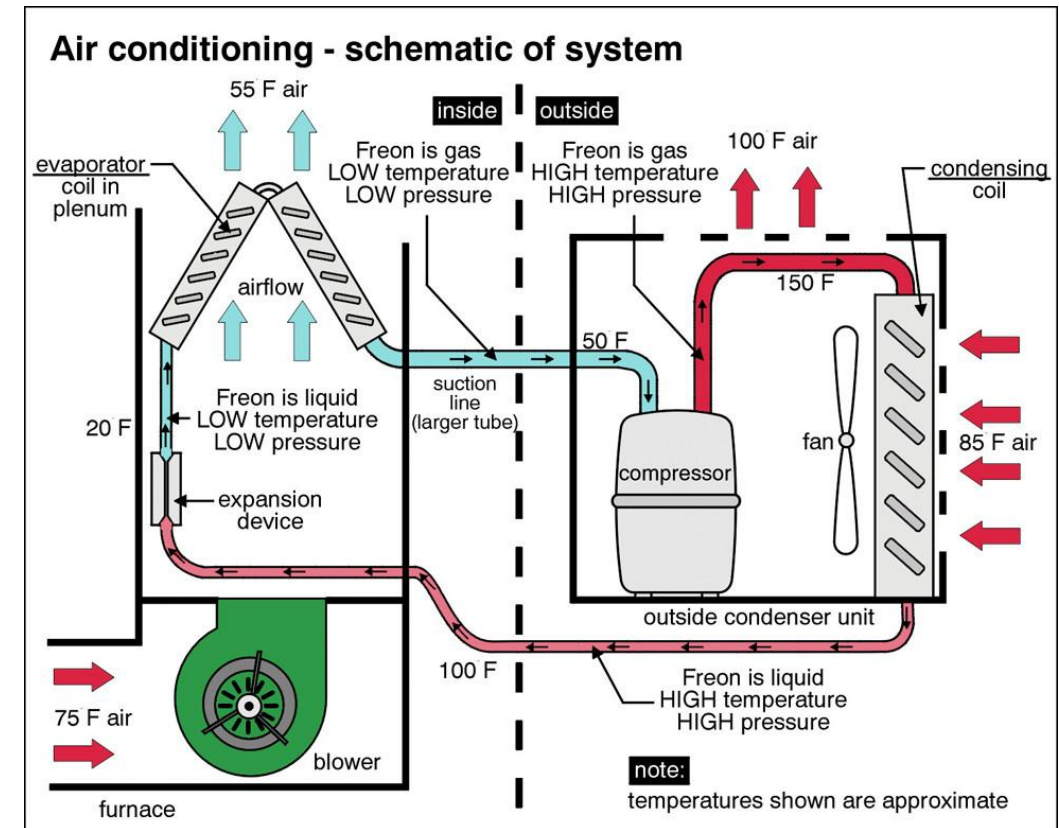
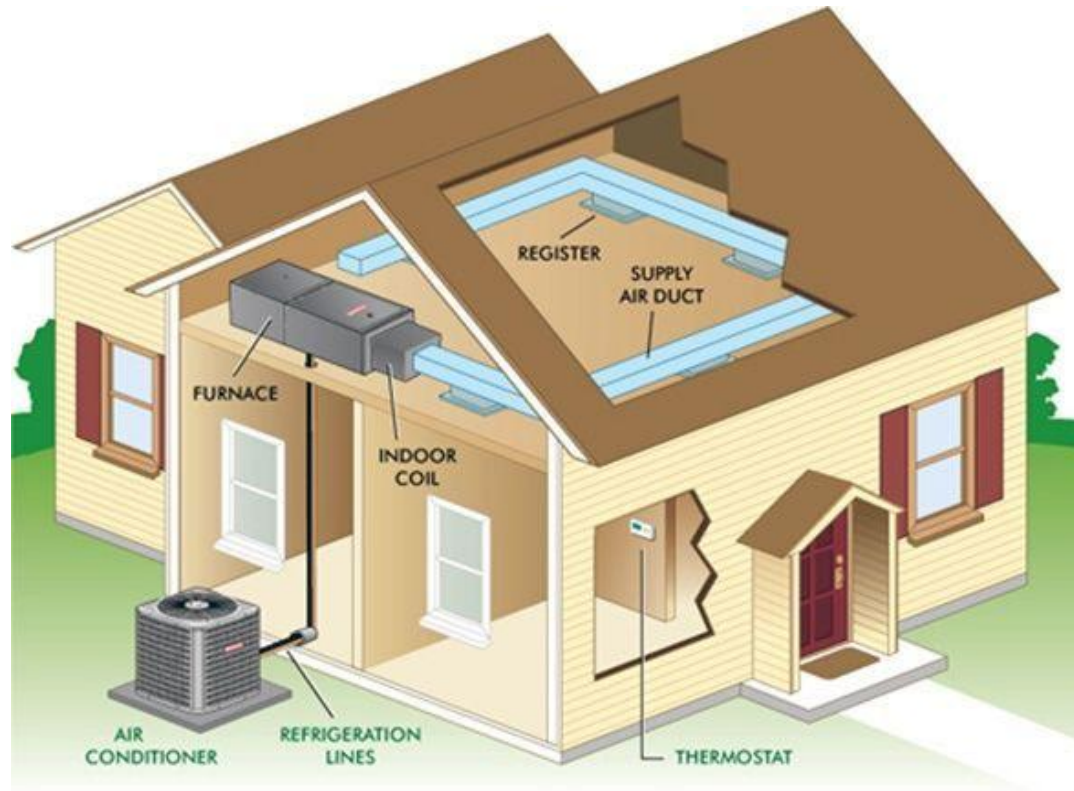
Classification

- **Primary**:- Refrigerants which directly take part in the refrigeration system
- **Secondary**:- Refrigerants which are first cooled by primary refrigerants and then used for cooling purposes



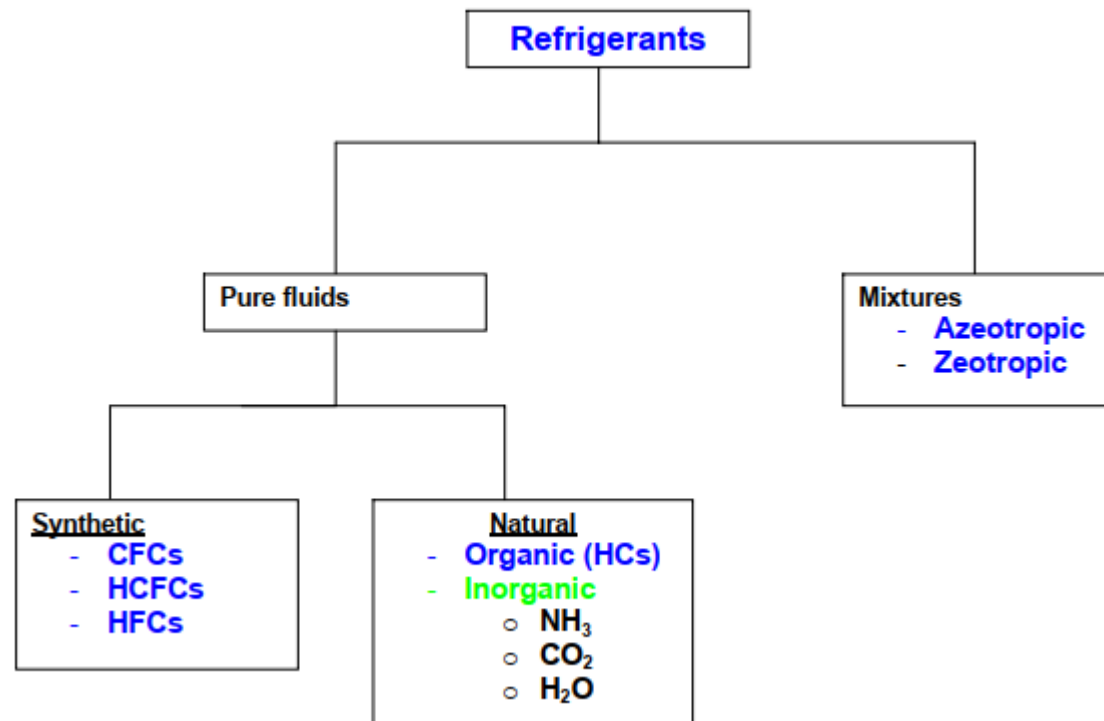
Example: Primary and Secondary Refrigeration Circuit

Example: Centralized AC



Primary Refrigerant: Freon
Secondary Refrigerant: Air

Types of Refrigerant



Designation of Refrigerants

1. Fully saturated, halogenated compounds:- Refrigerants that are derivative of alkanes (C_nH_{2n+2}) such as Methane/ethane etc. can be named as

- **R-XYZ**

- Where **X+1**: Number of carbon atoms
- **Y-1**:- Number of Hydrogen atoms
- **Z**: Number of Fluorine
- Balance indicates number of Chlorine atoms, Only two digits indicate that value of X is 0

Molecular formula	Non-proprietary name
CCl_3F	R11
CCl_2F_2	R12
$C_2H_3Cl_2F$	R141b
$C_2H_3ClF_2$	R142b
CHF_3	R23
C_3H_8	R290

Designation of Refrigerants

2. Inorganic Refrigerants: Designated by 7 followed by molecular weight of the refrigerant

- Example:

Refrigerant	Molecular Weight	Nomenclature
Ammonia	17	R717
Carbon Monoxide	44	R744
Water	18	R718

Designation of Refrigerants

3. Mixtures

iii) **Mixtures:** Azeotropic mixtures are designated by 500 series, where as zeotropic refrigerants (e.g. non-azeotropic mixtures) are designated by 400 series.

Azeotropic mixtures:

R 500: Mixture of R 12 (73.8 %) and R 152a (26.2%)

R 502: Mixture of R 22 (48.8 %) and R 115 (51.2%)

R503: Mixture of R 23 (40.1 %) and R 13 (59.9%)

R507A: Mixture of R 125 (50%) and R 143a (50%)

Zeotropic mixtures:

R404A : Mixture of R 125 (44%), R 143a (52%) and R 134a (4%)

R407A : Mixture of R 32 (20%), R 125 (40%) and R 134a (40%)

R407B : Mixture of R 32 (10%), R 125 (70%) and R 134a (20%)

R410A : Mixture of R 32 (50%) and R 125 (50%)

4. Hydrocarbons

Propane (C_3H_8) : R 290

n-butane (C_4H_{10}) : R 600

iso-butane (C_4H_{10}) : R 600a

Unsaturated Hydrocarbons: R1150 (C_2H_4)
R1270 (C_3H_6)

Desirable Properties

A. Thermodynamic properties

1. Low boiling point
2. High critical and saturation temperature
3. High latent heat of vaporisation
4. Positive pressure in evaporator and condenser
5. Low freezing point
6. Easy to liquefy at moderate pressure and temperature

B. Chemical properties

1. Non-toxic and No effect on quality of stored food
2. Non-flammable and non-explosive
3. Non-corrosive to metal
4. Chemical stability in reacting
5. Non-irritating and odourless

Desirable Properties



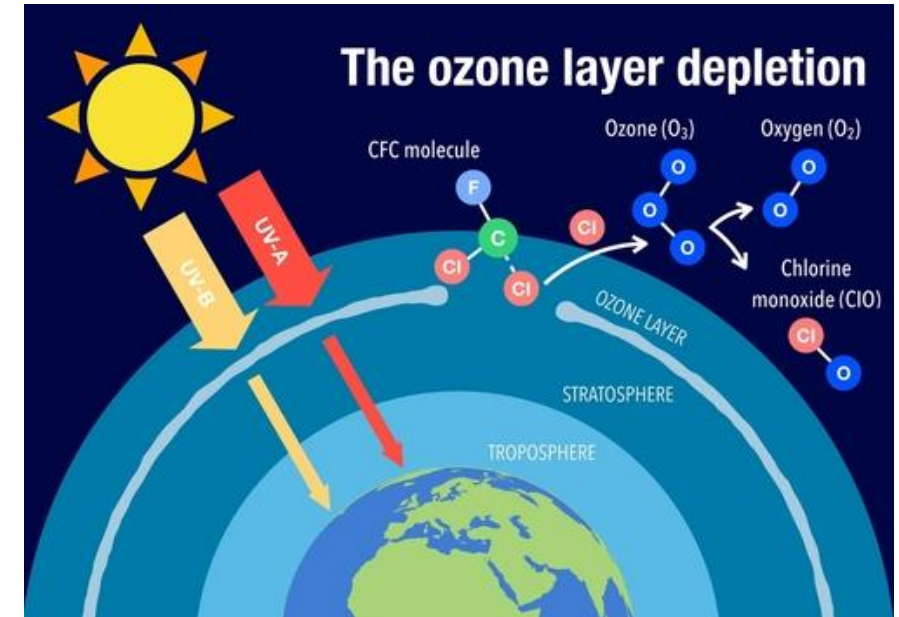
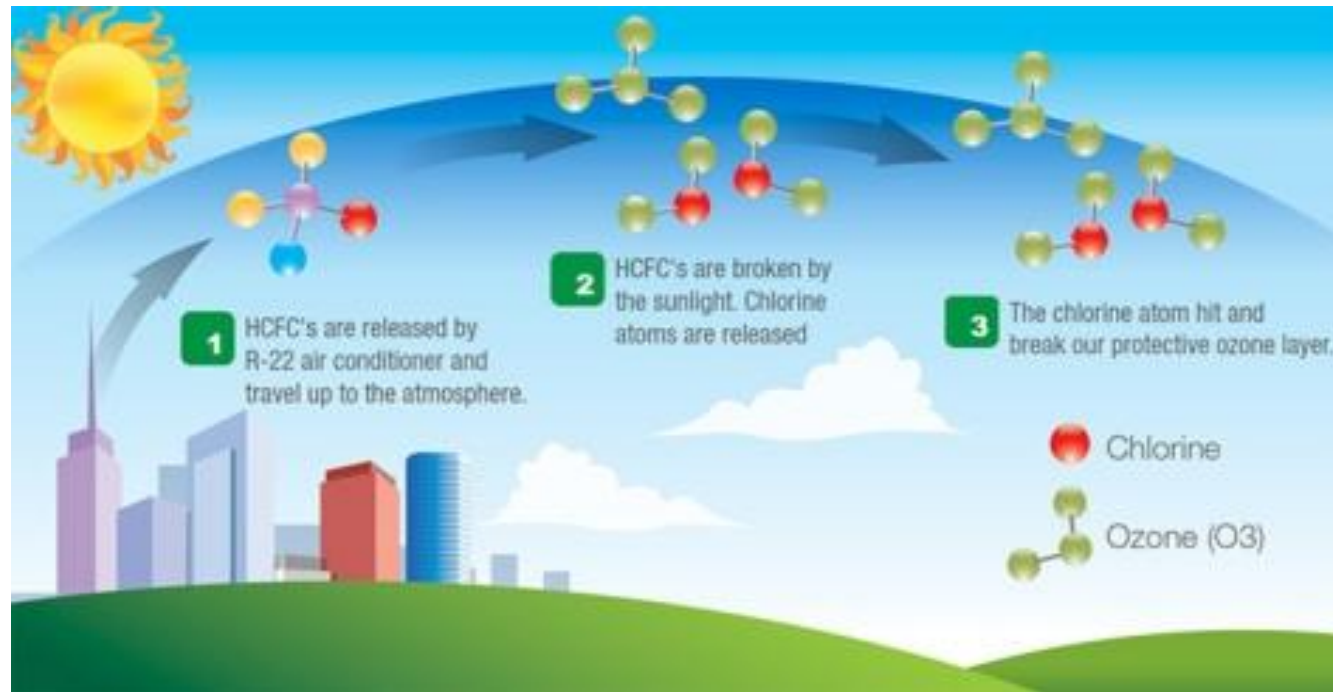
C. Physical properties

1. Low specific heat of liquid
2. Low specific volume of vapour
3. High thermal conductivity
4. Low viscosity
5. High electrical insulation

D. Other properties

1. Availability and Low cost
2. Easy of locating leaks by odour or suitable indicator
3. Mixes well with oil
4. Low power consumption
5. Low pressure ratio and High C.O.P

Environment Friendly Refrigerant



Environment Friendly Refrigerant

- Prior to the environmental issues of ozone layer depletion and global warming, the most widely used refrigerants were: R 11, R 12, R 22, R 502 and ammonia
- The depletion of stratospheric ozone layer was attributed to chlorine and bromine containing chemicals such as Halons, CFCs, HCFCs etc

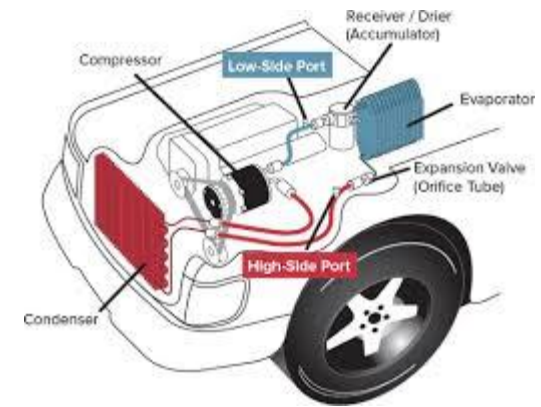
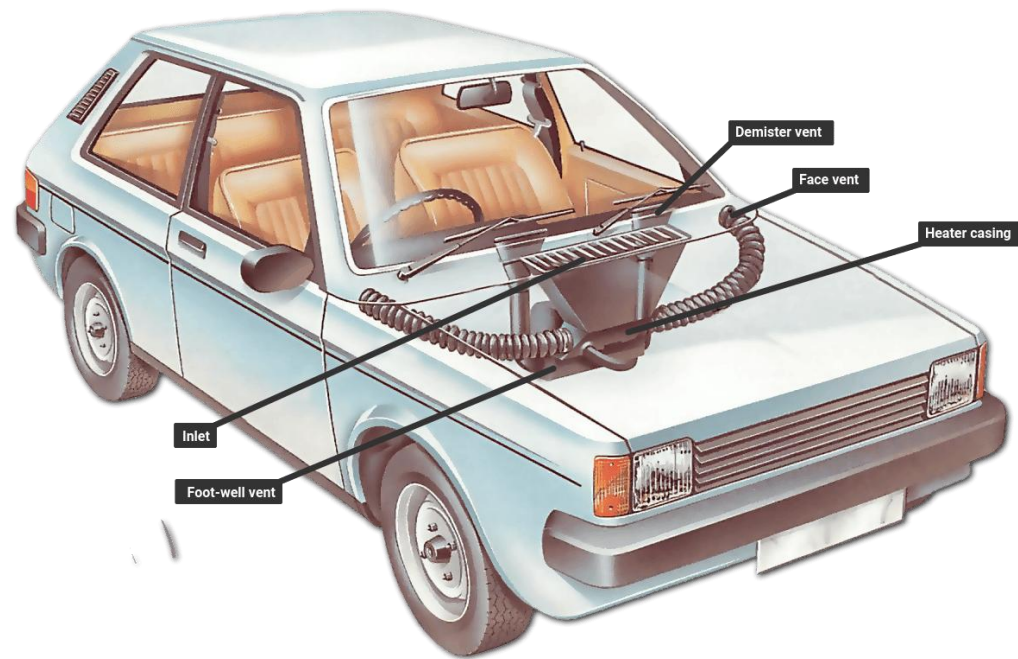
Refrigerant	Application	Substitute suggested Retrofit(R)/New (N)
R 11(CFC) NBP = 23.7°C h_{fg} at NBP=182.5 kJ/kg T_{cr} =197.98°C C_p/C_v = 1.13 ODP = 1.0 GWP = 3500	Large air conditioning systems Industrial heat pumps As foam blowing agent	R 123 (R,N)
		R 141b (N)
		R 245fa (N)
		n-pentane (R,N)
R 12 (CFC) NBP = -29.8°C h_{fg} at NBP=165.8 kJ/kg T_{cr} =112.04°C C_p/C_v = 1.126 ODP = 1.0 GWP = 7300	Domestic refrigerators Small air conditioners Water coolers Small cold storages	R 22 (R,N)
		R 134a (R,N)
		R 227ea (N)
		R 401A,R 401B (R,N)
		R 411A,R 411B (R,N)
		R 717 (N)
R 22 (HCFC) NBP = -40.8°C h_{fg} at NBP=233.2 kJ/kg T_{cr} =96.02°C C_p/C_v = 1.166 ODP = 0.05 GWP = 1500	Air conditioning systems Cold storages	R 410A, R 410B (N)
		R 417A (R,N)
		R 407C (R,N)
		R 507,R 507A (R,N)
		R 404A (R,N)
		R 717 (N)
R 134a (HFC) NBP = -26.15°C h_{fg} at NBP=222.5 kJ/kg T_{cr} =101.06°C C_p/C_v = 1.102 ODP = 0.0 GWP = 1200	Used as replacement for R 12 in domestic refrigerators, water coolers, automobile A/Cs etc	No replacement required * Immiscible in mineral oils * Highly hygroscopic
R 717 (NH₃) NBP = -33.35°C h_{fg} at NBP=1368.9 kJ/kg T_{cr} =133.0°C C_p/C_v = 1.31 ODP = 0.0 GWP = 0.0	Cold storages Ice plants Food processing Frozen food cabinets	No replacement required * Toxic and flammable * Incompatible with copper * Highly efficient * Inexpensive and available
R 744 (CO₂) NBP = -78.4°C h_{fg} at 40°C=321.3 kJ/kg T_{cr} =31.1°C C_p/C_v = 1.3 ODP = 0.0 GWP = 1.0	Cold storages Air conditioning systems Simultaneous cooling and heating (Transcritical cycle)	No replacement required * Very low critical temperature * Eco-friendly * Inexpensive and available

Table 26.1: Refrigerants, their applications and substitutes

Refrigerants and its applications

Refrigeration application	Short description	Typical HFCs used
Domestic Refrigeration	Appliances used for keeping food in dwelling units.	HFC-134a
Commercial Refrigeration	Holding and displaying frozen and fresh food in retail outlets	R 404A, R 507, HFC-134a
Food Processing and Cold Storage	Equipment to preserve, process and store food from its source to the wholesale distribution point	R410A, R407C, R 507, HFC-134a
Industrial Refrigeration	Large equipment, typically 25 kW to 30 MW, used for chemical processing, cold storage, food processing and district heating and cooling	HFC-134a, R-404A, R-507
Transport Refrigeration	Equipment to preserve and store goods, primarily foodstuffs, during	R410A, R407C, HFC-134a

HVAC System in Car



Primary Refrigerants

1. Halo-carbon Refrigerants:- refrigerants which contain one or more of three halogens, chlorine and bromine

Refrigerant number	Chemical name	Chemical formula
R-11	Trichloromonofluoromethane	CCl_3F
R-12	Dichlorodifluoromethane	CCl_2F_2
R-13	Monochlorotrifluoromethane	CClF_3
R-14	Carbontetrafluoride	CF_4
R-21	Dichloromonofluoromethane	CHCl_2F
R-22	Monochlorodifluoromethane	CHClF_2
R-30	Methylene chloride	CH_2Cl_2

Primary Refrigerants

2. Azeotrope refrigerants:- mixture of different substances. These substances cannot be separated into components by distillation.

- They possess fixed thermodynamic properties and do not undergo any separation with changes in temperature and pressure

Refrigerant number	Chemical name	Chemical formula
R-500	73.8% R-12 and 26.2% R-152	$\text{CCl}_2\text{F}_2/\text{CH}_3\text{CHF}_2$
R-502	48.8% R-22 and 51.2% R-115	$\text{CHClF}_2/\text{CClF}_2\text{CF}_3$
R-503	40.1% R-23 and 59.5% R-13	$\text{CHF}_3/\text{CClF}_3$
R-504	48.2% R-32 and 51.8% R-115	$\text{CH}_2\text{F}_2/\text{CClF}_2\text{CF}_3$

Primary Refrigerants



3. Inorganic refrigerants

Refrigerant number	Chemical name	Chemical formula
R-717	Ammonia	NH_3
R-718	Water	H_2O
R-729	Air	Mixture of $\text{O}_2, \text{N}_2, \text{CO}_2$ etc
R-744	Carbon dioxide	CO_2
R-764	Sulphur dioxide	SO_2

Primary Refrigerants



4. Hydro-carbon refrigerants

Refrigerant number	Chemical name	Chemical formula
R-50	Methane	CH_4
R-170	Ethane	C_2H_6
R-290	Propane	C_2H_8
R-600	Butane	C_4H_{10}
R-600 a	Isobutane	C_4H_{10}

Primary Refrigerants

5. Unsaturated organic compound:- The refrigerants belonging to this group possess ethylene or propylene as their constituents

Refrigerant number	Chemical name	Chemical formula
R-1120	Trichloroethylene	$C_3H_4Cl_3$
R-1130	Dichloroethylene	$C_2H_4Cl_2$
R-1150	Ethylene	C_2H_4
R-1270	Propylene	C_3H_6

Secondary Refrigerant

- Examples:- Brine
- Brines are secondary refrigerants and are generally used where temperatures are required to be maintained below the freezing point of water i.e. 0°C
- Salt is mixed in water, then the freezing temperature of the solution becomes lower than that of the water

Application Brine

1. Breweries
2. Chemical plants
3. Diaries
4. Food process
5. Ice-creams
6. Ice-plant
7. Meat packing
8. Skating ring
9. Special low temperature.

Thank You